

# MiniLM Lab

A small language model, measured end to end

Technical report / 25 September 2026 / Anvit Devadiga

[github.com/AnvitDevadiga/MiniLM-Lab](https://github.com/AnvitDevadiga/MiniLM-Lab)

**17.58**

held-out perplexity

**3.58x**

CPU cache speedup

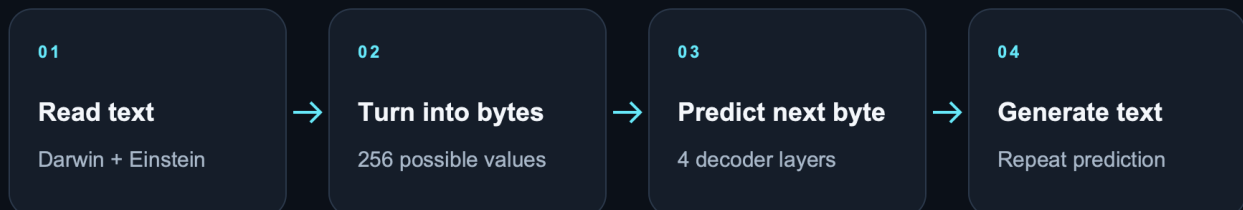
**53.6%**

stored-size reduction

## THE IDEA IN FOUR STEPS

### A tiny language model, end to end.

It learns which byte is likely to come next. Then it repeats.



Inside the decoder: causal attention · RMSNorm · SwiGLU · tied output weights

## Research question

Can a compact decoder-only language model learn held-out science prose while revealing measurable inference and storage trade-offs on laptop-class hardware?

Scope: one seed, approximately 1 MB of training text, CPU measurements. This is an educational systems study, not a general-purpose chatbot or a production deployment.

# What was trained and how?

## 01 Data and holdout

Public-domain editions of Darwin's *On the Origin of Species* and Einstein's *Relativity* were downloaded with pinned SHA-256 hashes. Gutenberg boilerplate was removed. Approximately the first 90% of each book was used for training; the remainder was held out at a paragraph boundary.

**1,008,977 training bytes**

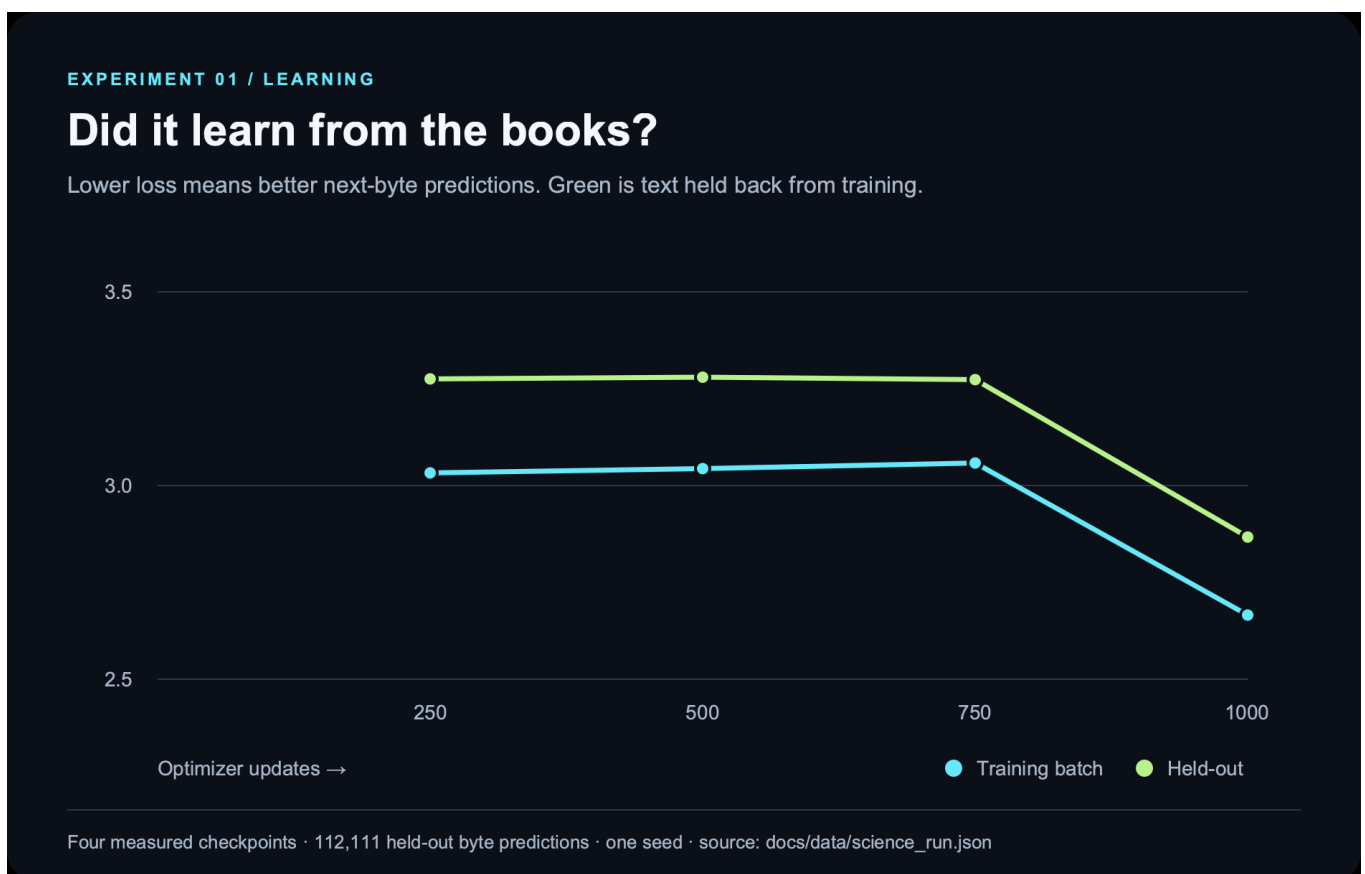
**112,112 held-out bytes**

## 02 Model and evaluation

Four decoder blocks; width 192; six heads; context 128; byte vocabulary 256; learned positions; RMSNorm; SwiGLU; tied embeddings. AdamW, learning rate  $3e-4$ , microbatch 8, four microbatches per optimizer update, seed 42. The best checkpoint is selected by deterministic full-split validation at updates 250, 500, 750 and 1,000.

Evaluation scores each next byte once in nonoverlapping 128-token chunks. This fixed-window, within-book holdout is not an unseen-author or knowledge benchmark.

## 03 Learning curve

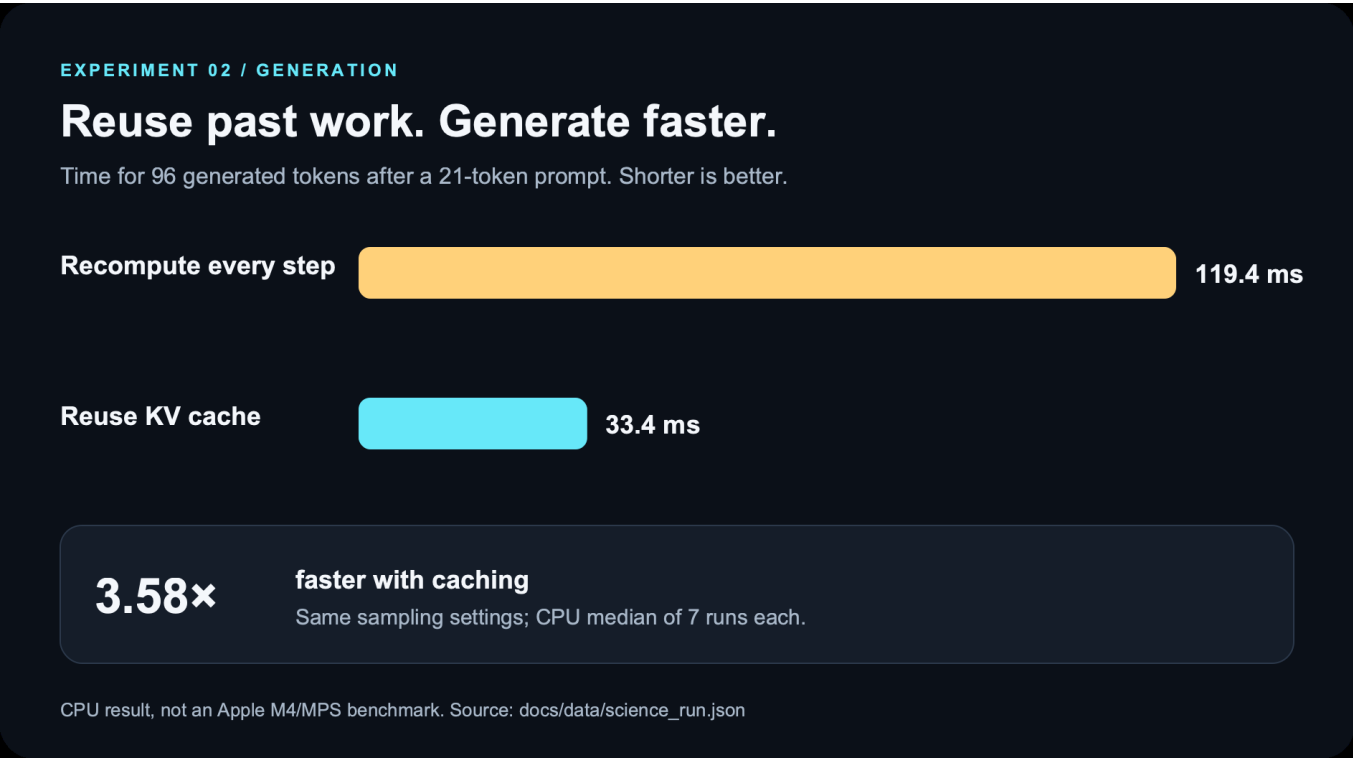


Final held-out loss: 2.8669 nats/byte; perplexity: 17.58; 112,111 scored targets. Four checkpoints are shown, not a continuous training trace.

# What changed in practice?

## 04 KV-cache latency

For a fixed 21-token prompt and 96 generated tokens, both modes used the same sampling settings. Each had one warm-up and seven timed CPU runs. Reported values are medians; lower latency is better.



## 05 Held-out results by book

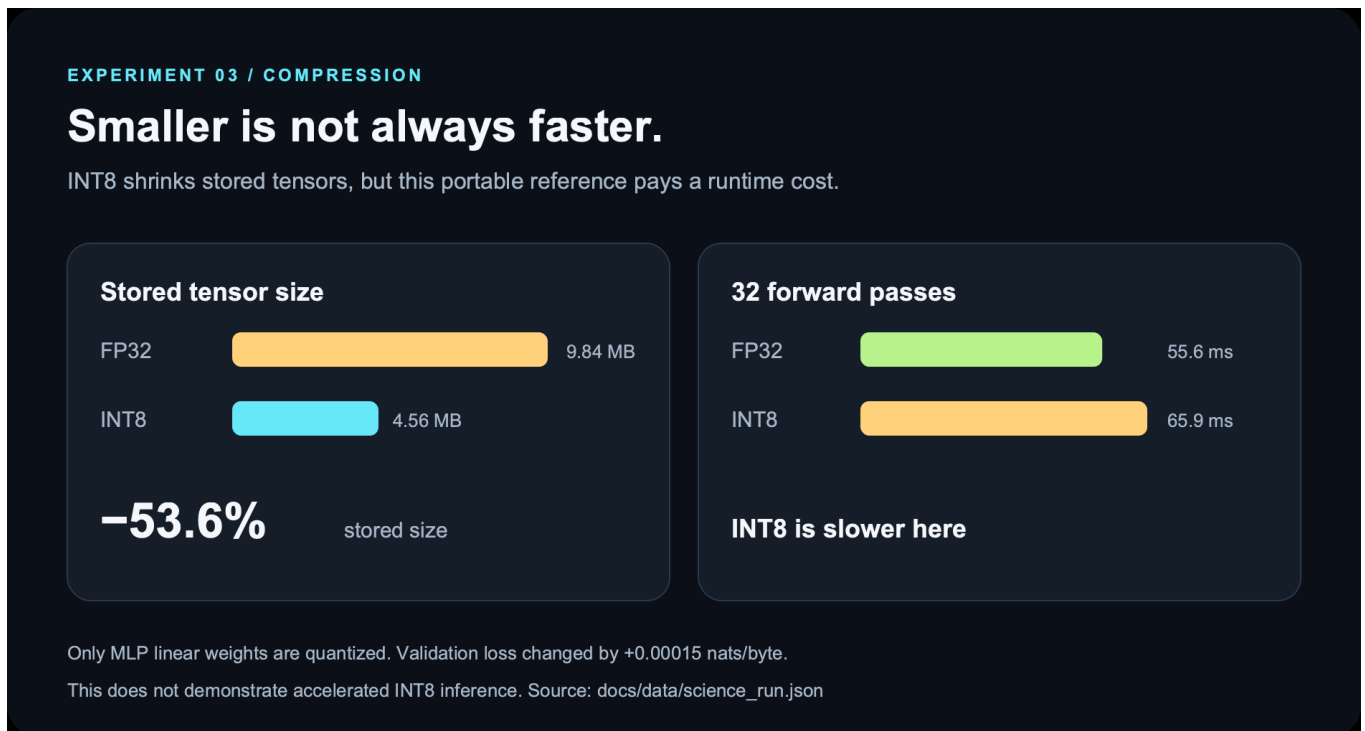
| Source   | Scored bytes | Perplexity | Bits/byte |
|----------|--------------|------------|-----------|
| Darwin   | 93,405       | 17.76      | 4.150     |
| Einstein | 18,703       | 16.68      | 4.060     |

Darwin contributes most of the held-out bytes. These are passages from the same books used for training, not a cross-book transfer test.

# The compression trade-off

## 06 What INT8 changes

Per-output-channel INT8 quantizes MLP linear weights only. Attention projections, embeddings and the tied output head remain FP32. This portable reference dequantizes weights during each forward pass.



## 07 Limits

This study uses one seed, one model size and approximately 1 MB of training text. Results have no confidence interval. The within-book holdout does not show scientific understanding, unseen-author transfer or general chatbot quality.

CPU timings are local to one execution environment. Apple M4/MPS throughput, peak memory, energy use and accelerated INT8 kernels were not measured. The result supports a storage reduction, not faster INT8 deployment.

## 08 Reproduce and inspect

The repository contains source hashes, prepared-corpus manifest, training and evaluation code, all benchmark samples, tests, and commands. The strongest next experiments are multiple seeds, a genuinely unseen-book holdout and a measured M4 run.

Code and protocol: [github.com/AnvitDevadiga/MiniLM-Lab](https://github.com/AnvitDevadiga/MiniLM-Lab)